Eat Well, Live Well.

AJINOMOTO

Growth Strategy for the Electronic Materials Business

Virtual Tour of Ajinomoto Fine-Techno Co., Inc.'s Gunma Plant

Genjin Mago

President and Representative Director
Ajinomoto Fine-Techno Co., Inc.

June 30, 2026

- Forward-looking statements, such as business performance forecasts, made in these materials are based on management's estimates, assumptions and projections at the time of publication. A number of factors could cause actual results to differ materially from expectations.

- This material includes summary figures that have not been audited so the numbers may change.

- “AminoScience” is a registered trademark of Ajinomoto Co., Inc.

AJINOMOTO

Today's Message

- **Since its launch, *ABF*™ has established an overwhelming position in its market, maintaining a share of over 95%. With our high-speed development system, we anticipate customer needs from the early stage of product development and are building a competitive advantage based on collaboration throughout the value chain.**

- **From this foundation, we are improving this product by co-creating value and enhancing performance in line with the evolution of semiconductors, and are capturing the growth of AI and HPC demand to accelerate growth.**

- **FY2026 has had strong performance from the start. There are no concerns about our current supply chain overall. Looking at the medium to long term, we will steadily expand our supply capacity and meet growing demand by establishing new bases aimed at 2030 and beyond, by developing new generation of *ABF*™.**

- **We will also anticipate and address the evolution of future technologies, such as co-packaged optics. In addition to evolving *ABF*™, we will extend its technology into areas such as next-generation packages to achieve sustainable growth.**

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

The Semiconductor Market

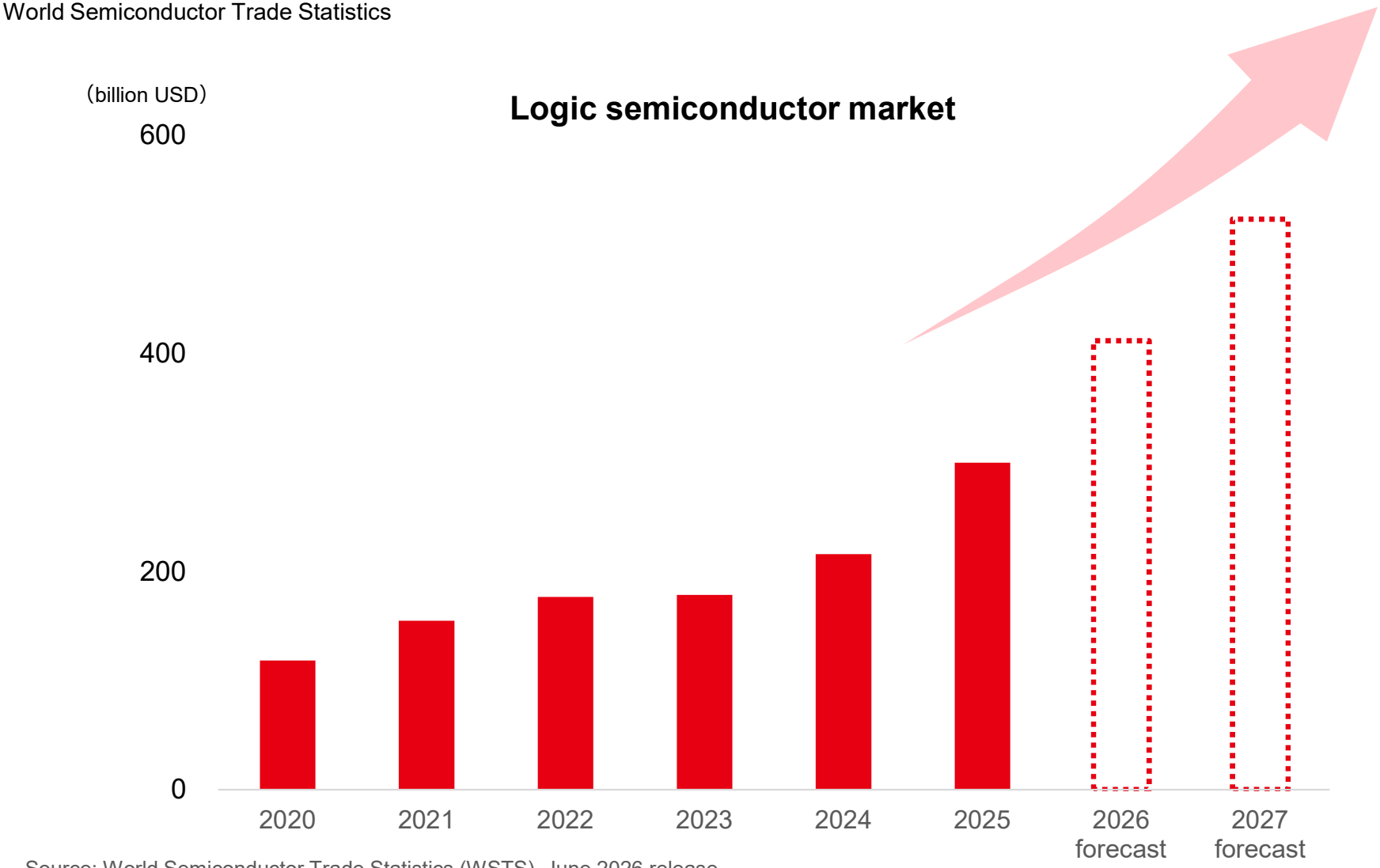
Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Eat Well, Live Well.
AJINOMOTO

Outlook for the Semiconductor Market (via WSTS)

In the logic IC market, WSTS* predicts strong growth of about 37% in 2026 and about 27% in 2027. Strong growth is expected to continue amid growing AI-related demand.

* World Semiconductor Trade Statistics



Source: World Semiconductor Trade Statistics (WSTS), June 2026 release

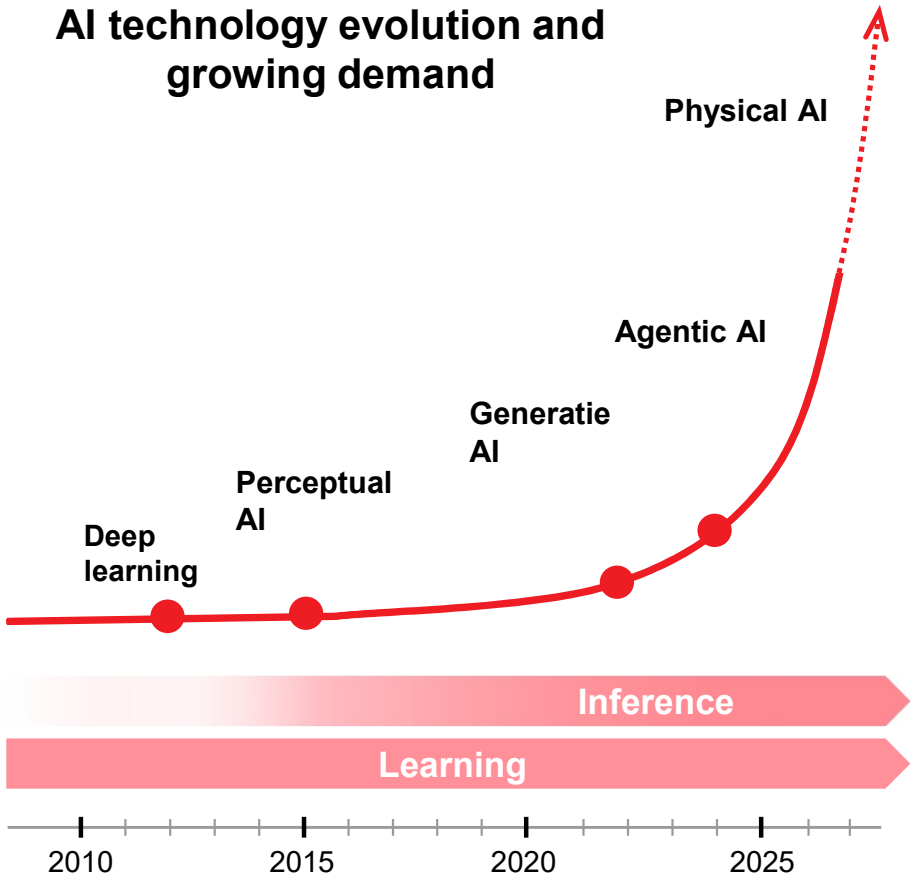
Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Eat Well, Live Well.
AJINOMOTO

The Evolution of AI Semiconductors: From Learning to Inference

Growth in the semiconductor market is being driven by AI-related demand. AI is advancing in scope from existing learning to inference, and the nature of processing is changing with a view toward the expansion of AI agents. Because of this, the configurations and roles of semiconductors supporting this change are also evolving.

AI technology evolution and growing demand



Changes in chip configuration and impacts on the market

The shift from learning to inference has changed the nature of processing, along with the configurations of chips supporting it.

- **Learning:**
  Calculation by large-scale models largely uses **high-performance GPUs**.

- **Inference:**
  With emphasis on response speed and power efficiency, the use of **CPUs and ASICs in addition to GPUs** is advancing.

⇒ Demand for package substrates is growing in terms of both quantity (increased chip count) and quality (high performance)

Source: Prepared by our company from various materials

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Our Electronic Materials Business, *ABF*™

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

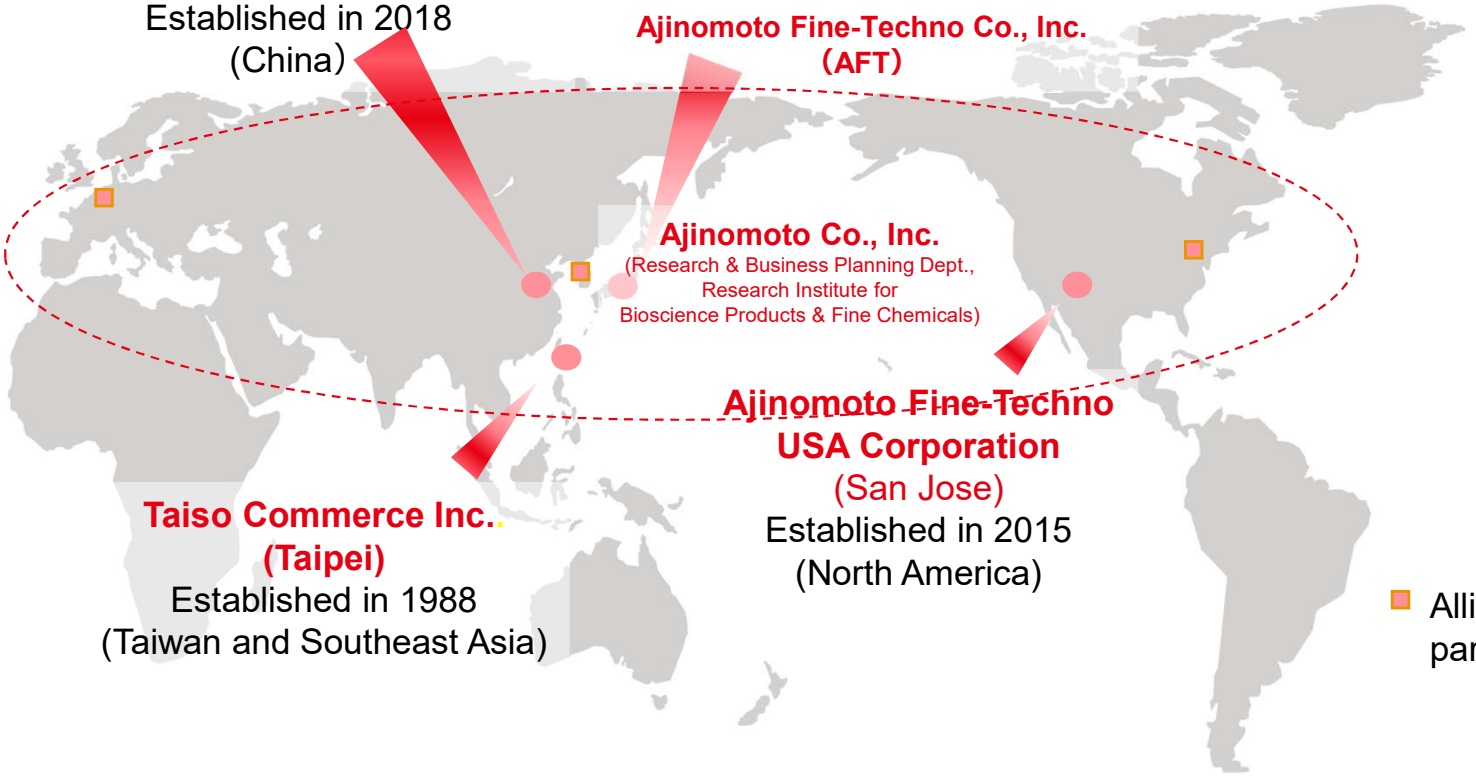
Eat Well, Live Well.
AJINOMOTO

Our Electronic Materials Business: Global Network

Our Electronic Materials Business operates globally, centered on Ajinomoto Fine-Techno Co., Inc. (AFT). The business supports the semiconductor development of customers worldwide.



A core company in the development, manufacture, and sales of electronic materials
Established: September, 1942
Capital: 315 million yen
Employees: 487 (as of April 1, 2026)
Sites: Kawasaki and Gunma



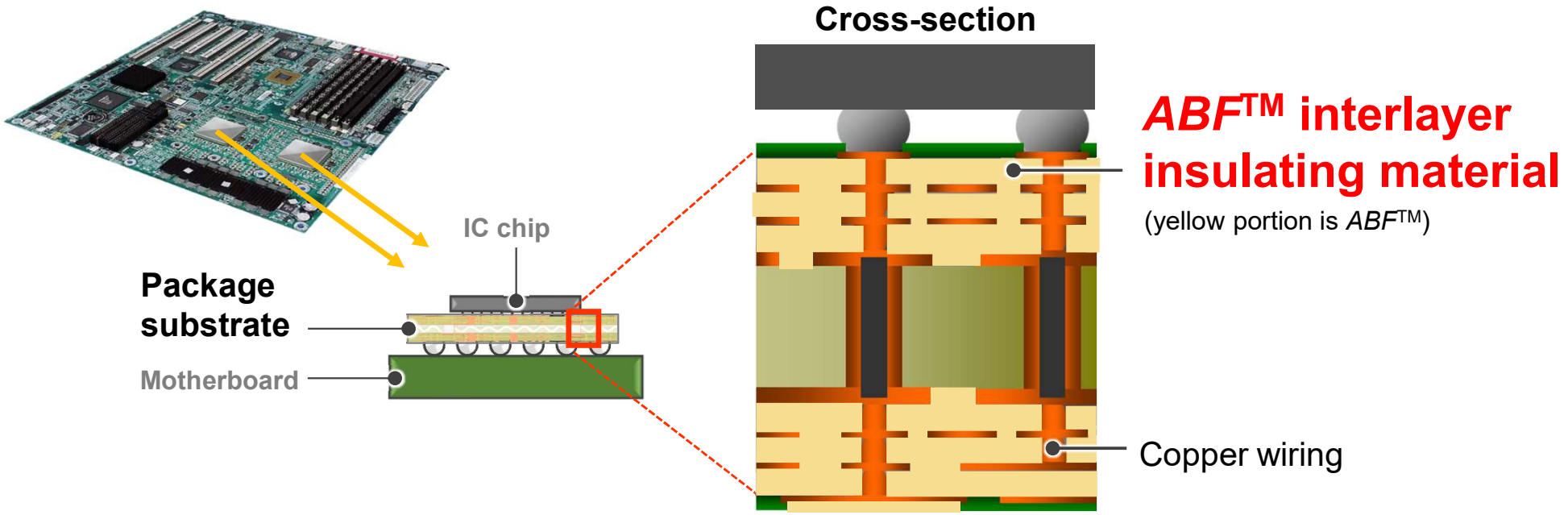
▶ **We build partnerships that support customers' medium- to long-term business growth through a global network of sites.**

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Eat Well, Live Well.
AJINOMOTO

ABF™: An Interlayer Insulating Material Supporting Logic ICs

The main product of our Electronic Materials Business, *ABF*™, is an insulating film used between layers of package substrates for logic ICs. *ABF*™ sees use in PCs, network equipment, servers, AI-related equipment, gaming devices, and more.



Main categories of logic ICs

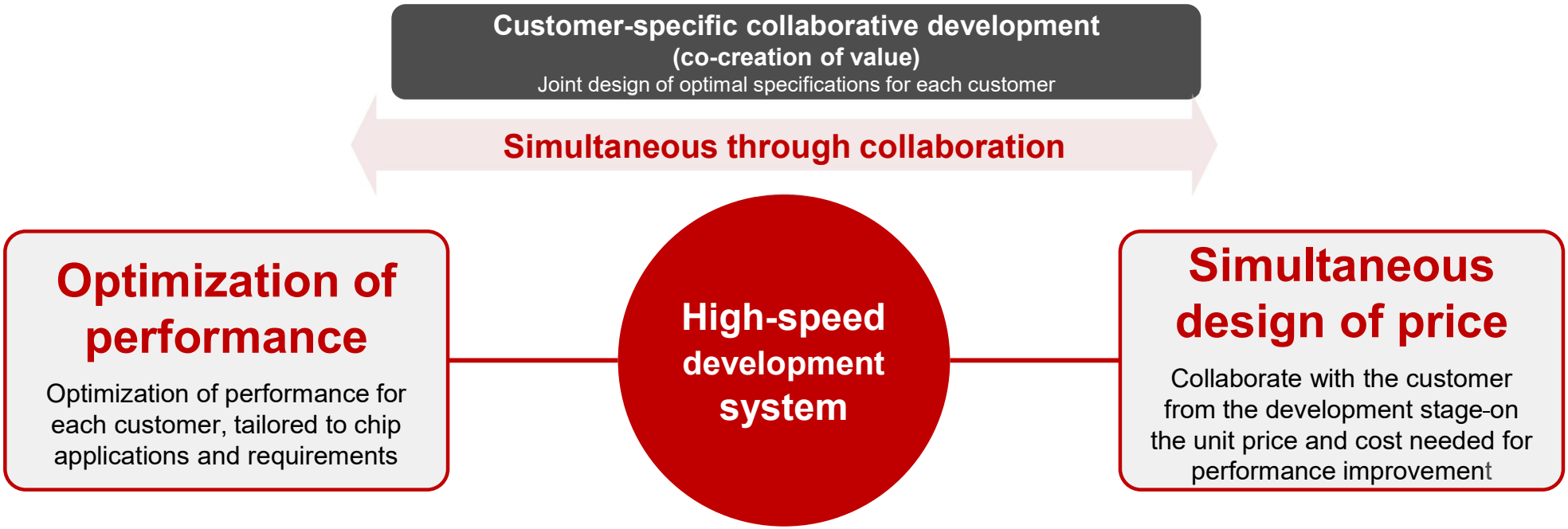
CPU Central Processing Unit	GPU Graphics Processing Unit	ASIC Application-Specific IC
A processor with advanced arithmetic processing functions, for use in servers and PCs	A parallel computing chip used in image processing and AI	ICs designed and manufactured for specific applications such as AI

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Eat Well, Live Well.
AJINOMOTO

The *ABF*™ Business Model: Customer-Specific Collaborative Development (Co-Creation of Value)

***ABF*™ is custom-manufactured for each customer. From the start of its development, *ABF*™ is designed to be optimal for the customer by simultaneously adjusting price and performance through collaborative development with them. We will continually provide *ABF*™ tailored to their needs.**



Simultaneous design of performance and price from the start of development, leading to sustainable provision of value and reliability

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Eat Well, Live Well.
AJINOMOTO

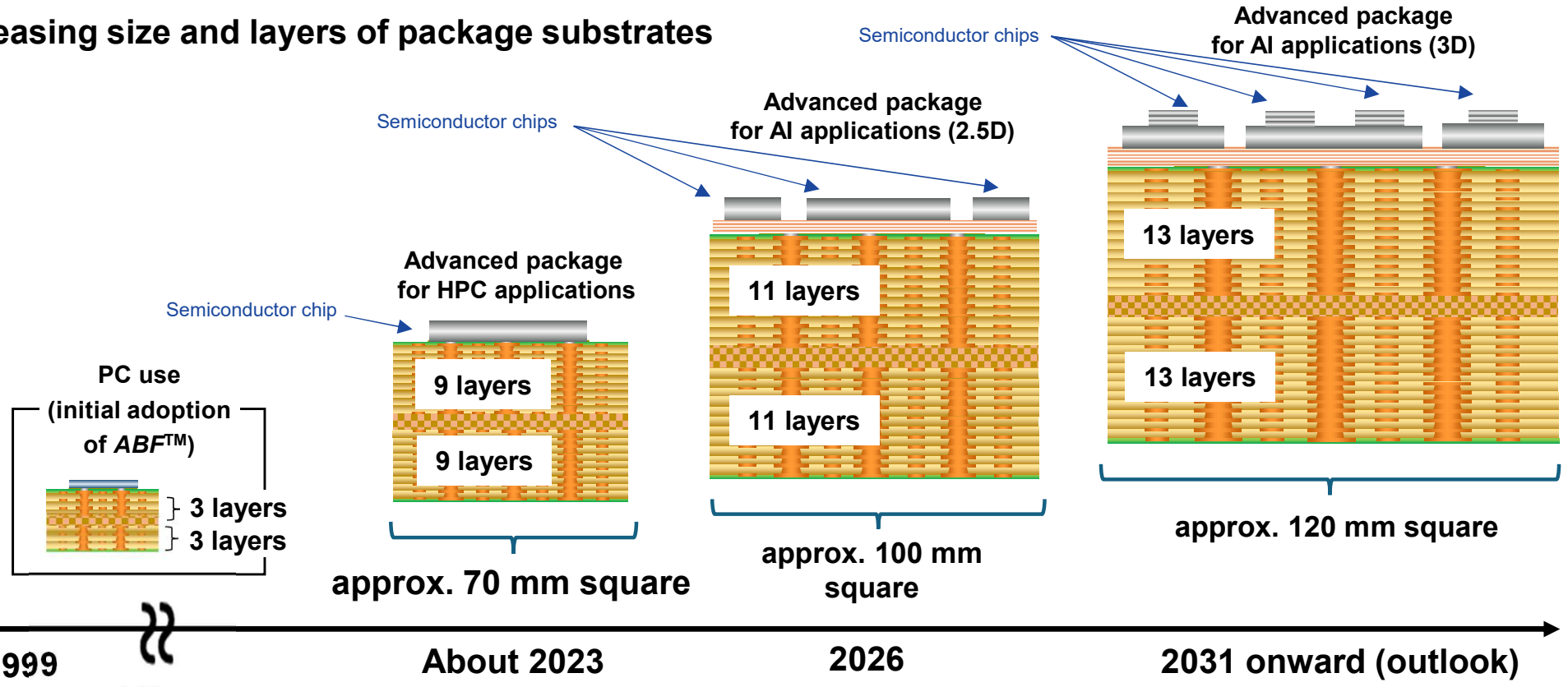
Increasing Size and Layering of Package Substrates and Increasing Amount of *ABF*™ Used per Substrate

As technology evolves in the ICT industry, semiconductors are becoming larger and having a higher level of integration, and package substrates are also becoming larger and having more layers. In line with this, the amount of *ABF*™ used for high-performance semiconductors is also increasing.

Evolution of the ICT industry

late 1980s onward	late 2000s to late 2010s	2020s onward	2031 (outlook)
Emergence of PCs, Internet	Emergence of mobile devices, cloud	Evolution of AI, data-driven technology	Further advance of AI

Increasing size and layers of package substrates



Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Eat Well, Live Well.
AJINOMOTO

ABF™ in AI Servers

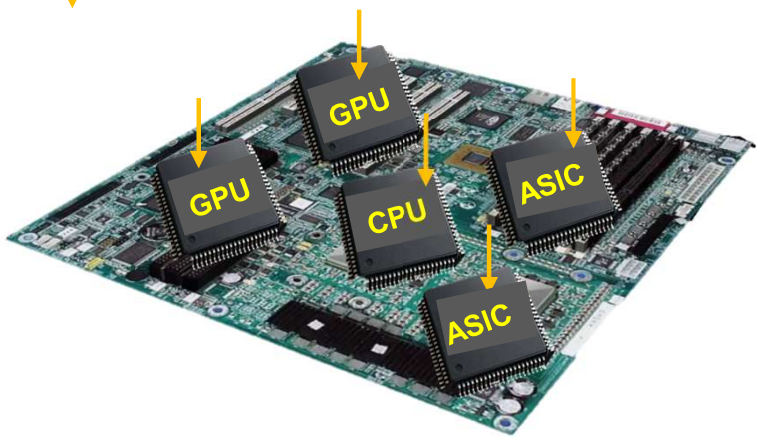
AI servers have large numbers of high-performance semiconductor packages, such as CPUs, GPUs, and ASICs. These large, multi-layer package substrates are of a construction that uses a large amount of *ABF*™.

The size and the number of layers of package substrates for high-performance semiconductors is increasing. The use of *ABF*™ is increasing.

(1) **Multiple high-performance semiconductors are combined and put on a system substrate***

*A substrate on which semiconductors such as the CPUs, GPUs, and ASICs are mounted

***ABF*™ is used in each package substrate**



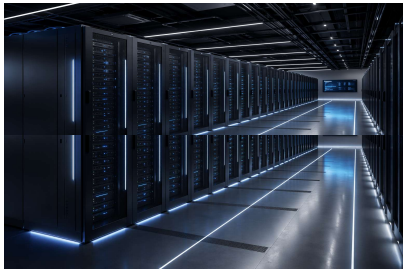
(2) **Combine multiple system substrates to form a high-performance "compute unit"****

(3) **Multiple compute units are combined to form an AI server**

AI server (representative image)



(4) **A number of servers are installed at a data center**



Hyper-scale data center (representative image)

A large amount of *ABF*™ is used

**Compute unit: A configuration in which multiple system substrates, memory modules, power supplies, and other components are combined into a single unit capable of independent calculations.

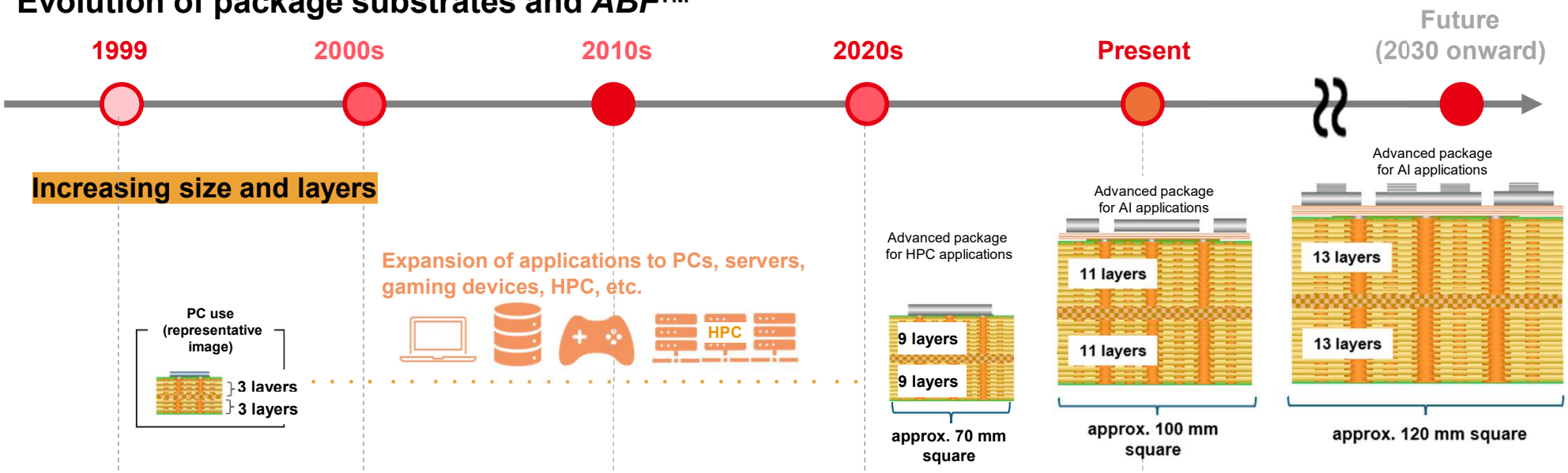
Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Eat Well, Live Well.
AJINOMOTO

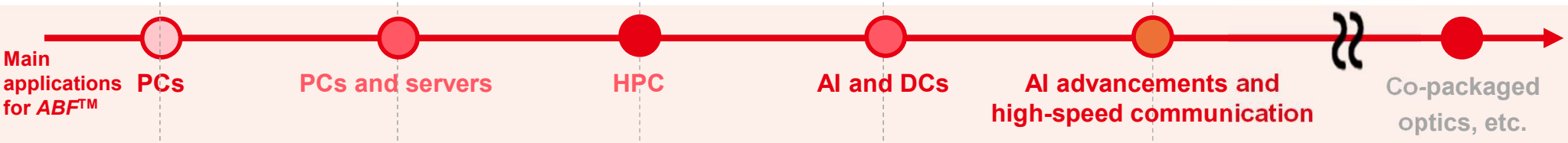
The Evolution of *ABF*™ in Step with the Evolution of Package Substrates

We continue to evolve *ABF*™, achieving performance in step with the technological evolution of package substrates. Package substrates are expected to become larger and have more layers with the evolution of AI. We are pursuing the advancement of *ABF*™ technology to meet this change.

Evolution of package substrates and *ABF*™



Evolution of *ABF*™



1999

Development of the world's first build-up material in film form

Improved mounting through use of film
Secured consistent thermal resistance and reliability

2000s–2010s

Compatible with miniaturization and high reliability

Improved surface smoothness and resistance to heat, and compatible with micro-wiring

2020s onward

Compatible with miniaturization, high-speed transmission, and low loss

Existing functions plus compatible with advanced processes and brought about reduced transmission loss

Present and future

Compatible with increasingly large size and more layers

Compatible with high-bandwidth, large-scale I/O; brought about both reliability (warping, humidity) and reduced transmission loss, etc.

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Eat Well, Live Well.
AJINOMOTO

Changes in the *ABF*™ Product Lineup (representative image)

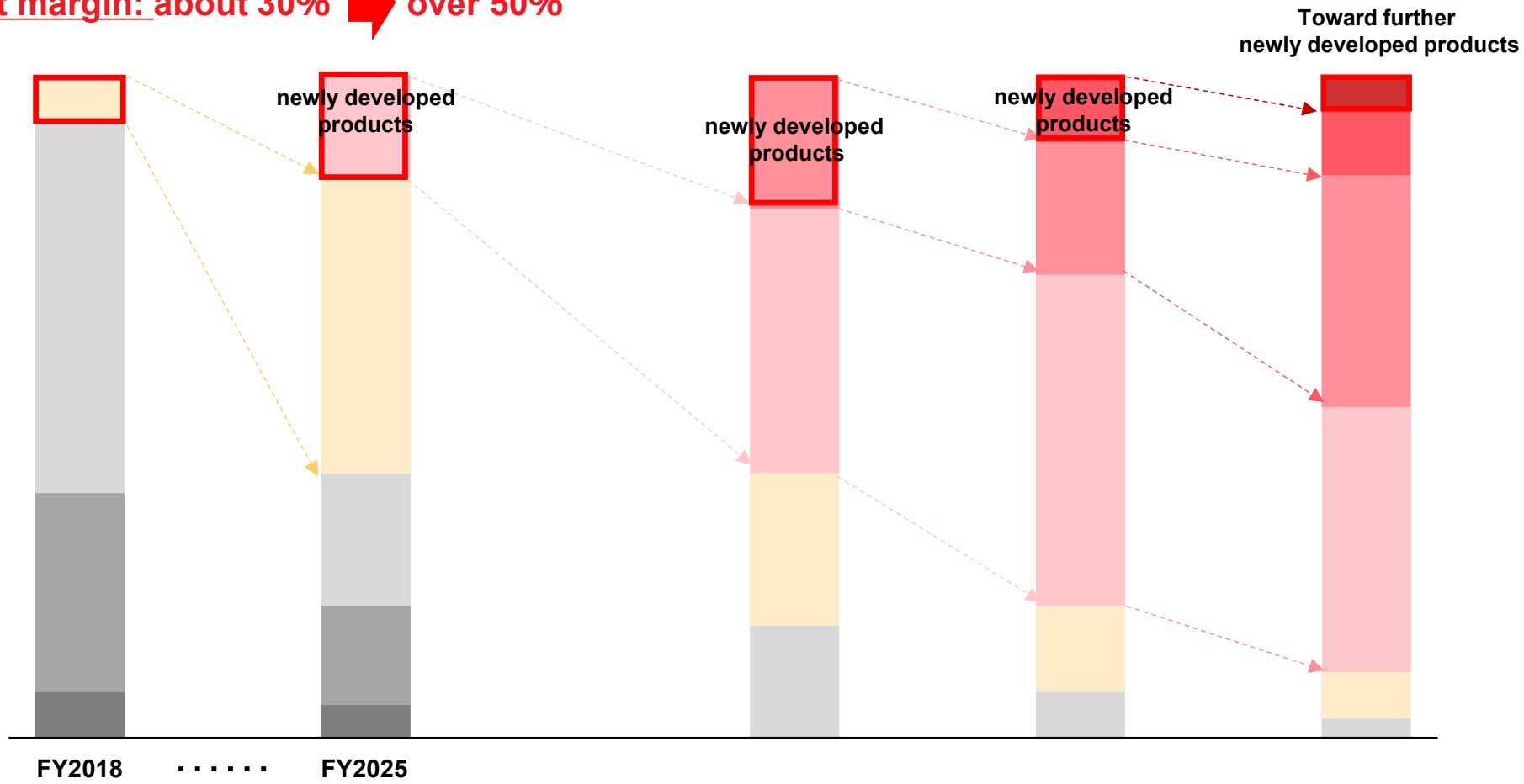
We develop new and higher value-added products, and the increasing share of these products in our portfolio has contributed to higher profit margins. Going forward, we will also continue to develop new and higher value-added products.

Changes in the *ABF*™ product lineup

Representative image of how the product lineup has changed

Representative image of the future product lineup

Profit margin: about 30% → over 50%



Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

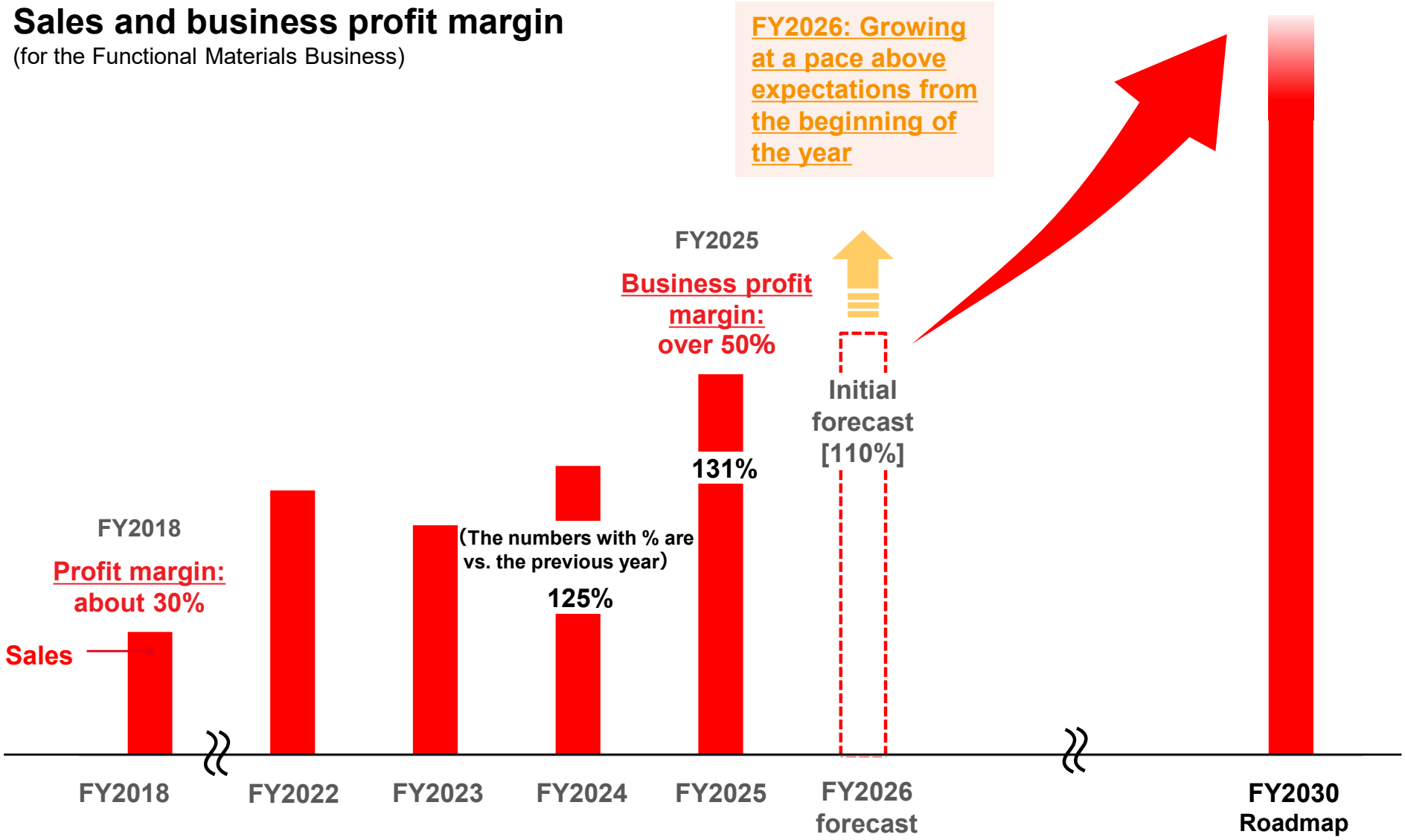
Eat Well, Live Well.
AJINOMOTO

Outlook for Growth up to FY2030

Driven by strong demand for *ABF*™ in leading-edge fields, performance has been robust since the start of FY2026. Looking ahead to FY2030, too, we expect dramatic growth in *ABF*™ sales due to increased volume and higher value-added products, supported by our stable supply system.

Sales and business profit margin

(for the Functional Materials Business)



Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

ABF™ Development and Production Structure

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Eat Well, Live Well.

The Mechanism Supporting the Evolution of *ABF*™: Our High-Speed Development System

Our high-speed development system anticipates future customer needs to always meet those needs in advance. We work closely with the players in the value chain to provide value with a unified system, from development to mass production.

Customers (package substrate manufacturers)

Feedback and evaluation

Provide value with high-speed development

Ajinomoto Group

The players in the value chain

- Raw material manufacturers
- Equipment manufacturers
- Chemical solution manufacturers
- End users, etc.

We have strengths in the co-creation of value with the players in the value chain, as well as with customers

Research departments

- Researchers directly communicate with customers
- Analytical skills developed through foods are used
- MI* is used to design materials at high speed.

Technology departments

- Small-lot prototype lines and evaluation equipment equivalent to the customers'
- Verification aimed at scaling up based on quick customer feedback

Production departments

- Stable quality and rapid transition to mass production
- Detailed technical support for mass-produced products

Management departments

- Quality assurance and quality control
- Build a global SCM and stable supply structure
- Assure compatibility with environmental, chemical substance, and other regulations

Sales departments

- Address customers' issues and build positive relationships over the medium to long term
- Propose new businesses and products

Ajinomoto Group Synergy:

Analysis and quality assurance technologies, developed through "AminoScience," function as the basis for the evaluation of materials for electronic materials

* Materials informatics, a technology that applies data science, AI, and machine learning methods to the development of materials

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

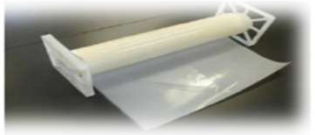
Eat Well, Live Well.
AJINOMOTO

Reproduction of Customer Processes to Support High-Speed Development

Substrate design premised on *ABF*™ has become standard due to our high-speed development, which is grounded in our track record and reliability. Also, our structure that reproduces customers' manufacturing and evaluation processes in our Group is a key foundation supporting high-speed development.

Steps in *ABF*™ development



Develop in the laboratory ▶ Make prototypes ▶ Have the customer evaluate the product ▶ Start mass production

High-speed development system

Customers' manufacturing and evaluation process

Fine-line patterning process

1 Embed → 2 Via formation process → 3 Copper wiring process → 4 Evaluate reliability → 5 Inspect for warpage → Completion


Vacuum laminate machine


CO_2 laser (form vias)


Lithography system (resist)


Reliability test equipment


Substrate warpage measuring device

▶▶**Our in-house reproduction of the fine-line process in particular is a key foundation that supports compatibility with the increasing size and layers of substrates**

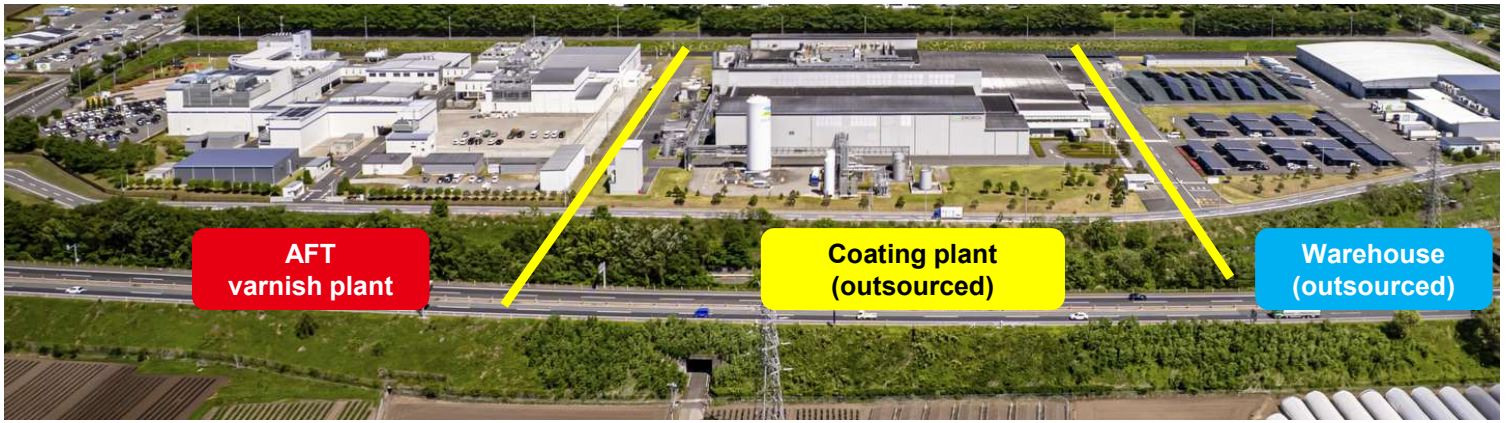
Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Eat Well, Live Well.
AJINOMOTO

The *ABF*™ Production Structure

We specialize in the production of varnish, the raw material for *ABF*™. Application to film and storage are outsourced. We are building a structure to provide a stable supply throughout the supply chain to meet the growing demand.

Bird's-eye view of the Gunma Plant

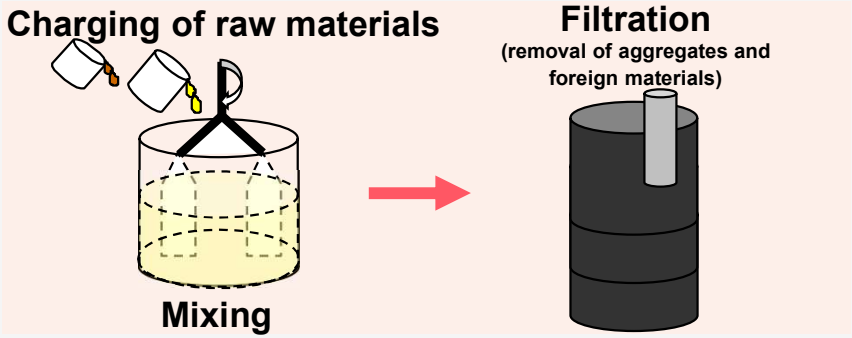


Steps in manufacturing

STEP 1 Procure raw materials ▶ STEP 2 Produce varnish ▶ STEP 3 Coat and cut ▶ STEP 4 Store in warehouse and ship

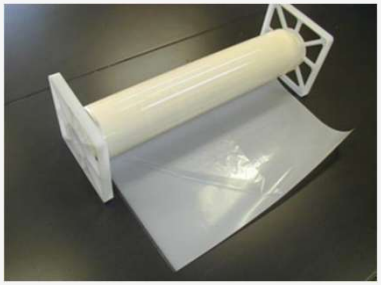
In-house manufacture

- Manufacture uniform varnish through precision weighing and homogenization



Outsourcing

- Varnish coating on PET film
- Store in freezer, ship domestically and internationally





Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Eat Well, Live Well.
AJINOMOTO

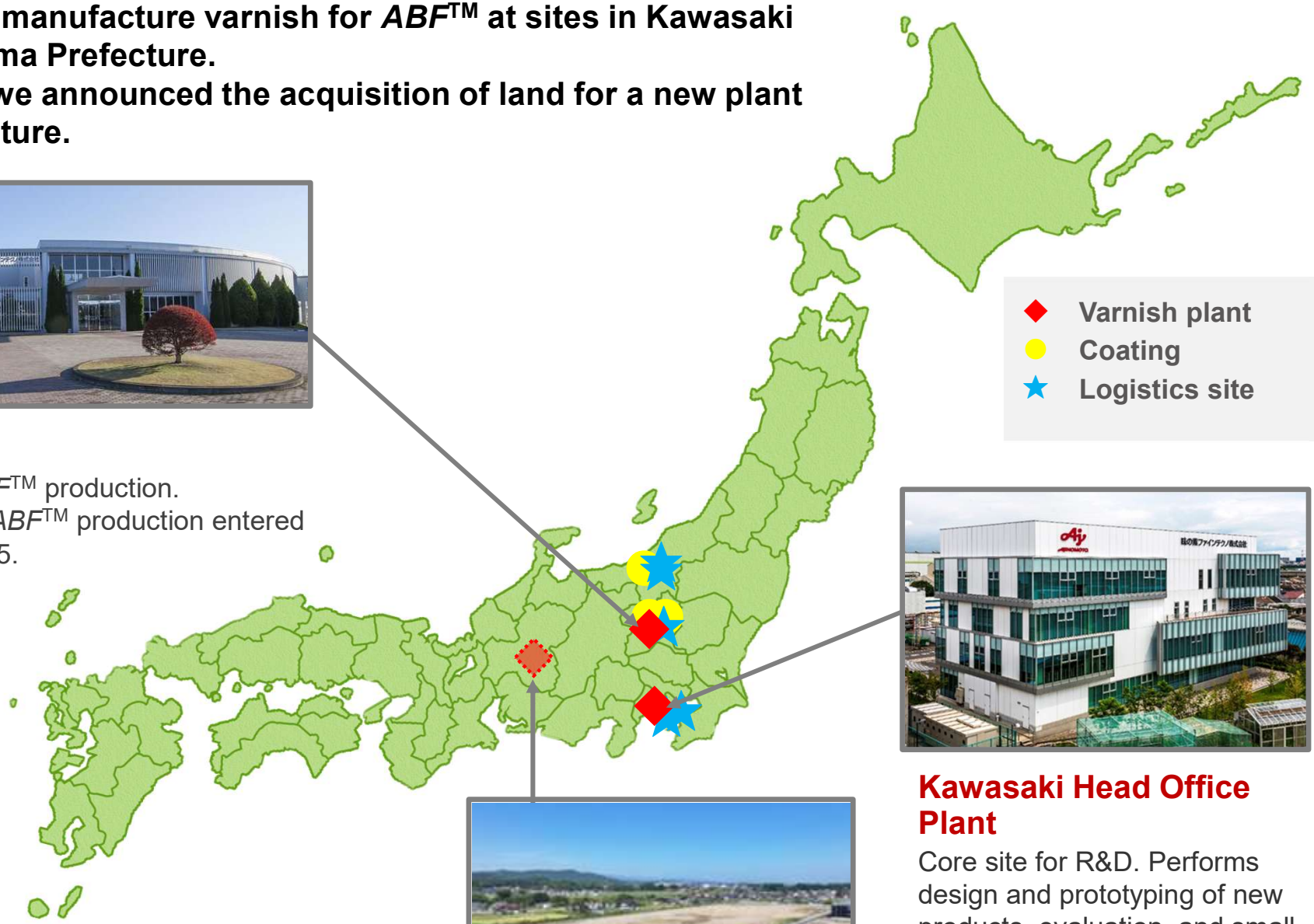
ABF™ Manufacturing Sites

Currently, we manufacture varnish for *ABF*™ at sites in Kawasaki City and Gunma Prefecture.
In May 2026, we announced the acquisition of land for a new plant in Gifu Prefecture.



Gunma Plant

Core site for *ABF*™ production.
A new plant for *ABF*™ production entered operation in 2025.



Kawasaki Head Office Plant

Core site for R&D. Performs design and prototyping of new products, evaluation, and small-lot production.



Gifu Plant
(operation scheduled for 2032)

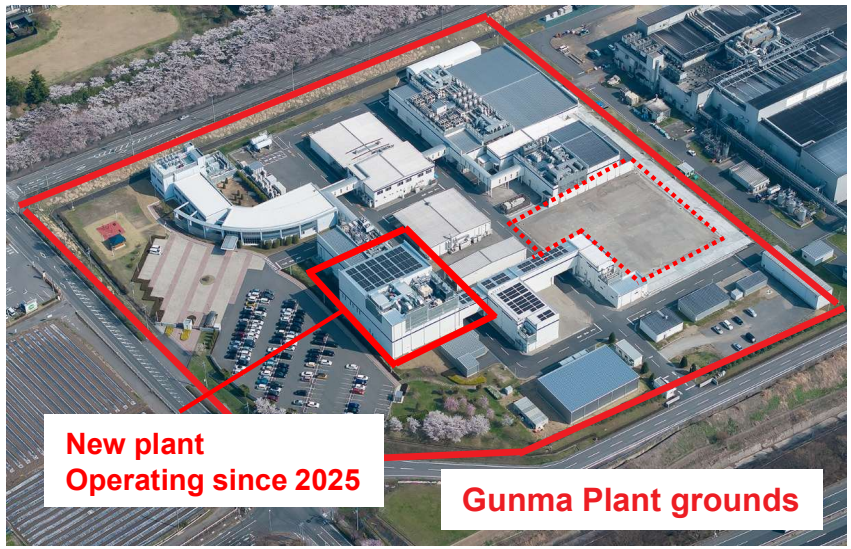
Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Eat Well, Live Well.
AJINOMOTO

ABF™ Manufacturing Sites (Development of Structure for Future Production Increases)

Anticipating growing demand through 2030, we will develop a structure for staged increases in production at the Gunma Plant and other locations.
To address increases in demand after 2030, we will develop a structure for further increasing production, including the newly acquired Gifu site, and build a comprehensive structure capable of fully meeting demand.

Structure for production increase through 2030



Gunma Plant

Current site for mass production of varnish for *ABF*™.
In 2025, we invested about ¥10 billion to establish a new plant for state-of-the-art varnish manufacturing.

Investment budget: Set to ¥25 billion between 2023 and 2030.

Structure for production increase after 2030



**Site for a plant
in Kani City, Gifu Prefecture**

Third production site after Kawasaki and Gunma.
The same production capacity as the Gunma Plant is expected through the introduction of DX.
Will also contribute to our BCP and risk diversification.

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Eat Well, Live Well.
AJINOMOTO

The AFT Gunma Plant: A New Plant with the Latest Equipment

The Gunma Plant is our main production base for *ABF*™, and the new plant will be compatible with the production of varnish for leading edge fields.
By introducing technologies for automation and to reduce the workforce needed, we are building a production system that combines high quality with high efficiency.



New Gunma Plant
(Full operation since 2025)

Features of the new plant

- Has state-of-the-art equipment for mass production of *ABF*™ varnish for state-of-the-art applications.
- Has IoT and automated conveying systems, promoting automation and the reduction of the workforce in the production process. This has brought about stable quality and enhanced production efficiency.

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

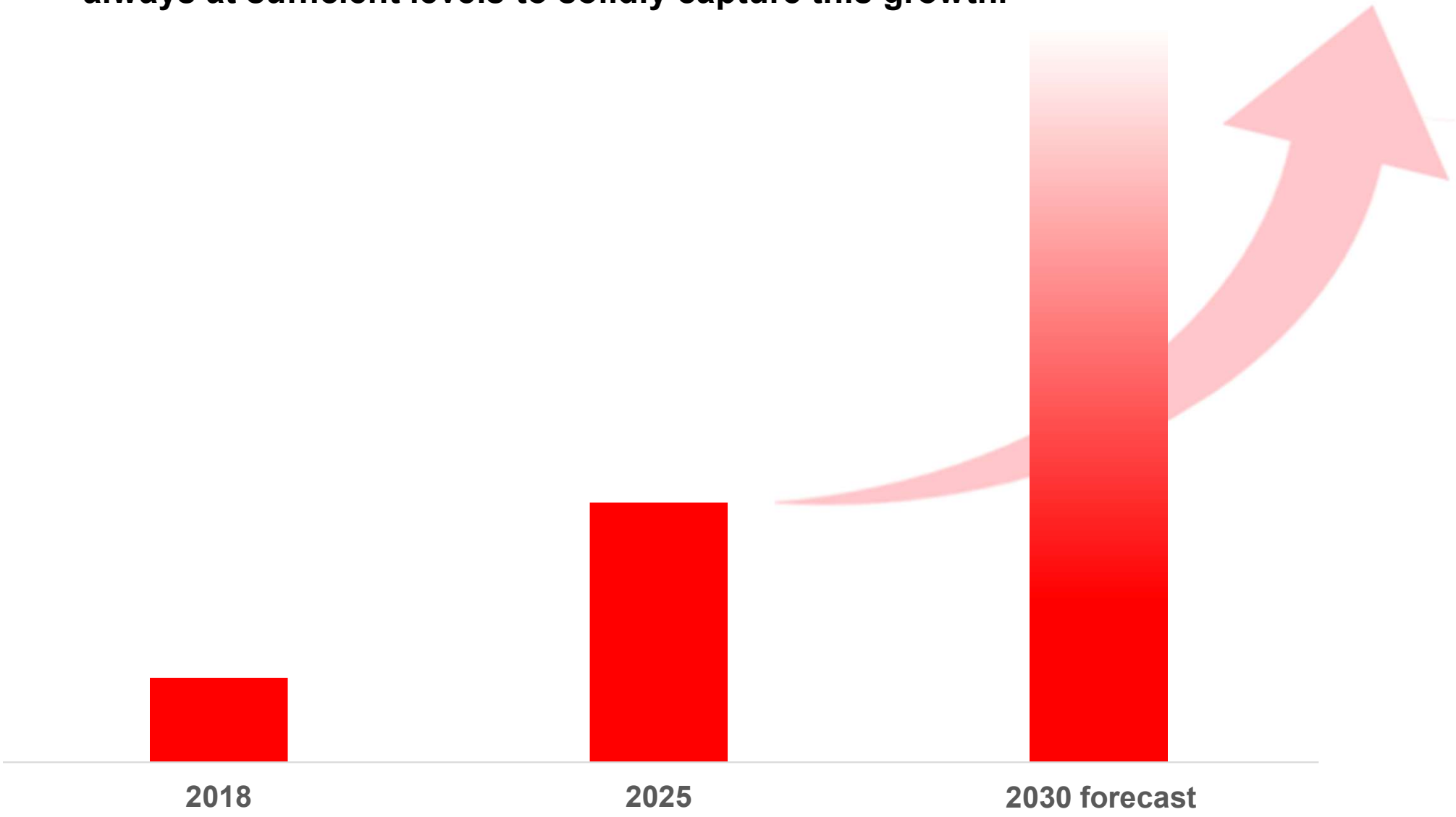


▶ Virtual tour of the new Gunma Plant manufacturing floor

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Functional Materials Business Sales Forecast

We expect the Functional Materials Business to grow dramatically heading toward 2030. We will invest strategically to ensure that production and supply capacity are always at sufficient levels to solidly capture this growth.



Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Addressing *ABF*™ Peripheral Areas and Future Technologies

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Extension of the Business to *ABF*™ Peripheral Areas

Building on technologies cultivated through *ABF*™, we are advancing the development and application of peripheral materials.

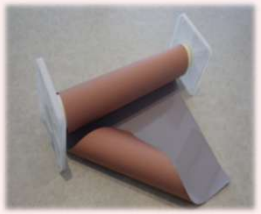
Our products

Adhesive materials



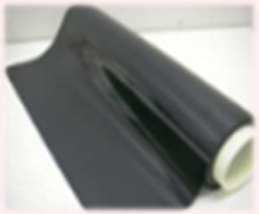
- Quick curing at low temperature
- High adhesion and flexibility

ABF™-RCC[1]



- Integration of copper foil into *ABF*™, contributing to higher performance and miniaturization

Molding materials



- Suppresses thermal deformation with a low CTE[2]
- Reduces warping and cracking

Magnetic materials



- Availabile in a variety of forms (film, paste, liquid)
- Contributes to high efficiency and power saving through low loss[3] and high adhesion

Applications (examples)

Electronic components, mobile devices (CCMs,[4] etc.)	**HDI substrates [5]**	**Protection, molding, and embedding of IC chips and components**	**Semiconductors for PCs, servers, 5G base stations, etc.**

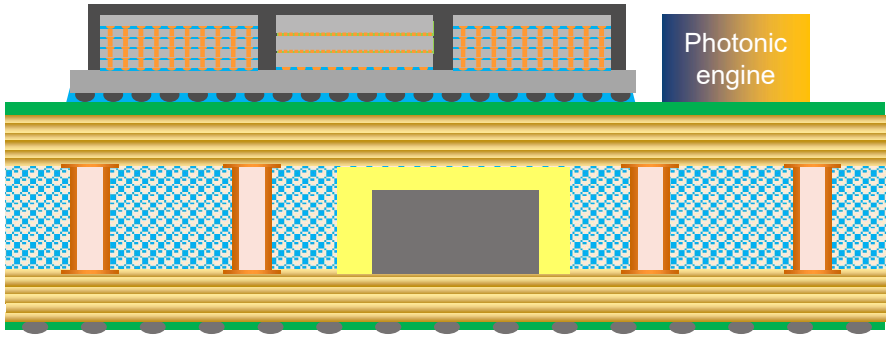
1: Resin Coated Copper foil
2: Low coefficient of thermal expansion: the property of resistance to expansion and contraction due to temperature changes, such as heating and cooling.
3: The property of low energy loss in the form of escaping heat when electrical signals or energy pass through a circuit or substrate
4: Compact camera modules
5: High Density Interconnect (HDI) substrates

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Addressing Next-Generation Packaging Technologies (The Future of Our Business with Technological Progress)

For next-generation packages that address future technological developments, we will not only make use of *ABF*™ but also our strengths, such as *ABF*™ technologies and infrastructure, to advance the development of materials that support requirements for electrical power, heat, bandwidth, power supply, connectivity, etc., as we take on the responsibility of making de facto standards.

Next-generation semiconductor package (example)



- Further increases in size and layers
- Use of optical signals as well as electrical signals
- Embedded component; further integration
- Use of high-performance substrate materials

Our business expansion potential (examples)

Arrows (⇒) indicate materials that we think we can provide

Increased size and layers
⇒ Multilayer substrate materials

- Technological evolution of *ABF*™
- Adaptation to new core materials; process materials

Electrical interconnect loss and thermal limitations
⇒ CPO (co-packaged optics) materials

- Evolution toward optical-guiding materials based on *ABF*™ technologies (optical waveguides, etc.)

Heterogeneous integration
⇒ Next-generation advanced heterogeneous packaging* materials

*Mechanism for operating varied types of chips/components as a single system

- Extension of *ABF*™ technology to redistribution layer and molding materials

We will develop business grounded in our core technologies and continue adapting to new technologies

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

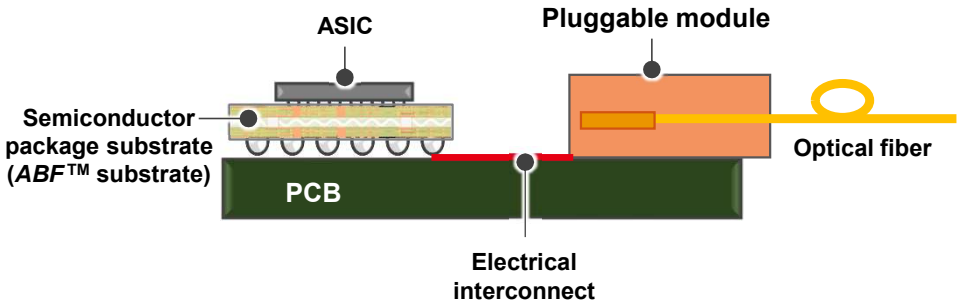
Addressing Next-Generation Package Technologies (Evolution from the Current State)

With the spread of AI and the resulting surge in the amount of data volumes and demand for communications, controlling power consumption in electrical signal-based high-speed transmissions has become an issue. To address this, the mounting of optical interfaces* near ASICs is attracting attention as one possibility.

*Interface for converting electrical signals into optical signals for transmission

Current state

- **Pluggable Optics: In data centers (particularly for AI and HPC), electrical interconnect power consumption increases as transmission speeds rise. To address this, technology that converts part of electrical signals into optical signals has been adopted.**

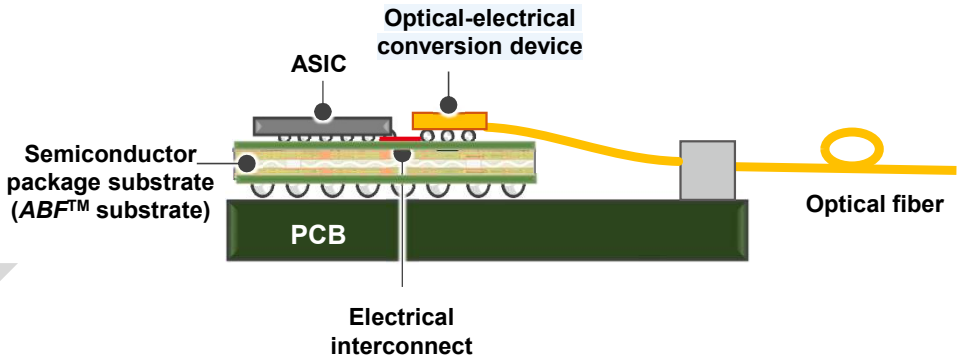


- The package substrate and ASIC are on the PCB substrate. An optical fiber connects from the pluggable module. Electrical signals travel between the ASIC and the pluggable module via the PCB.
- With faster speeds, loss from electrical interconnect and increased power consumption become issues.

≈

Future

- **CPO : A technology that integrates an optical interface into the same package as the ASIC to minimize electrical interconnect distances, significantly improving power efficiency and speed.**



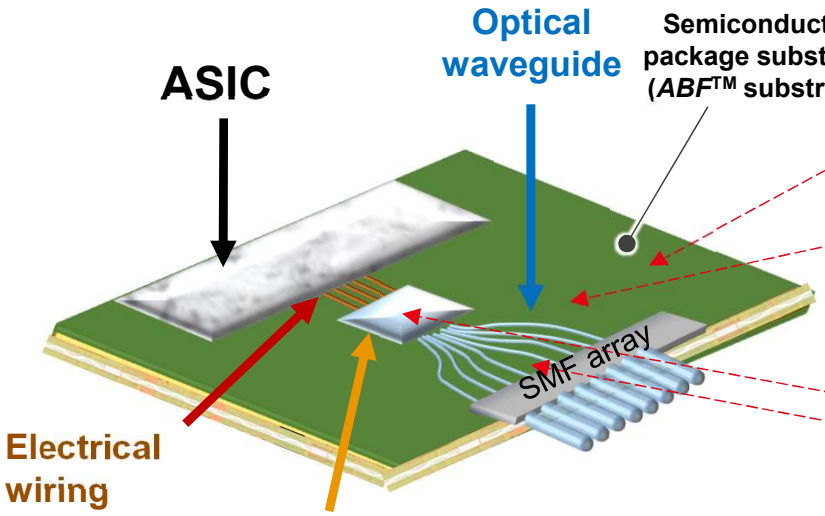
▶▶ See details on the next page

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Addressing Next-Generation Package Technologies (Evolution from the Current State)

CPO is a technology that performs optical-electrical conversion near ASICs in semiconductor packages to reduce power consumption by shortening the electrical interconnect distance. We will evolve *ABF*™ and leverage our strengths, such as our technologies and infrastructure, built up through *ABF*™, to contribute to the advancement of packages.

CPO substrate



Our business expansion potential (examples)

ABF™

Optical waveguides

- Polymer optical waveguide materials based on formulation design technology
- Design of means of guiding light (path, branching, loss)

Adhesives and molding materials

- For bonding optical engine components and fiber connections.
- Compensation for supporting alignment tolerance and stable optical coupling

- The semiconductor package and ASIC are placed on the PCB substrate. The configuration integrates the optical engine that performs optical-electrical conversion into the same package, and directly connects the semiconductor package to the external optical fiber.
- By significantly shortening the electrical interconnect distance between the ASIC and the optical engine and converting the electrical signal to an optical signal at an early stage, this achieves low power consumption and high-speed transmission.

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Organizational & Corporate Culture and Human Resources

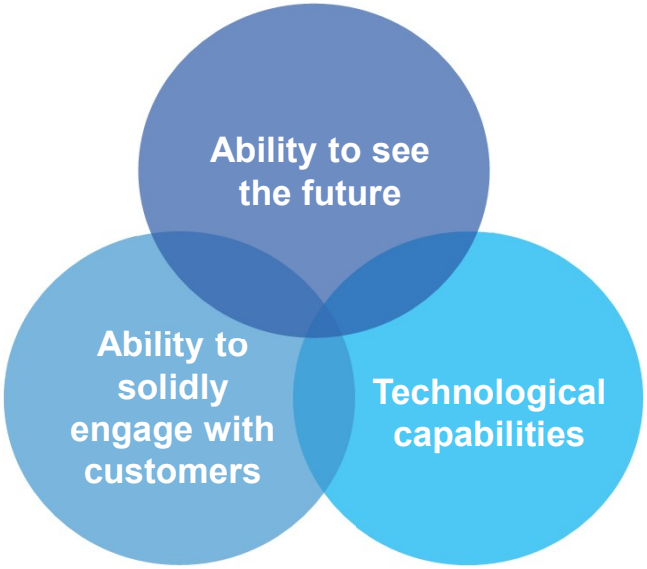
Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Our Vision for the Company and Our Sources of Competitiveness

To realize our vision, we will hone our core competencies, solidly engage with customers, continue tackling challenges, and always provide new value.

Our vision:
“Create comfort in living through unseen materials”

Core competency: A source of competitiveness that cannot be imitated by competitors
“Win before the competition even begins”



Ability to see the future
Actively collect comprehensive information from various sources

Ability to solidly engage with customers
Always deliver varied forms of value from AFT to customers

Technological capabilities
Advanced technological capabilities are a critical tool for delivering value

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Human Resources that Support Our High-Speed Development System

To strengthen research, technology development, manufacturing, quality assurance, sales, and other aspects of our high-speed development system, and to strengthen our sustainable relationships with customers, human resources in all of our departments collaborate globally to advance our business.



- **Our technical professionals own individual themes from the start and actively engage with customers and partners to understand on-site needs and challenges.**
- **We foster a culture that emphasizes quality throughout the company, with all employees proactively involved as parties concerned. We achieve high-quality products and services across departments under a culture of exchanging ideas for improvement regardless of position or seniority.**
- **Through customer-focused human resource development and cross-departmental collaboration, we reinforce our organizational capabilities so that we can continue to meet high-level requirements.**

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Organizational Culture and Talent Supporting the Competitiveness of the Electronic Materials Business

Eat Well, Live Well.
AJINOMOTO

What We Value

Face People with Integrity.
Engage sincerely with customers and colleagues. Look beyond the immediate issues in front of us, and keep our focus on people and the future ahead.

Explore the Unknown.
Continue to explore the unseen value of materials. Do not be constrained by what is currently visible; believe in our own capabilities to create new markets and generate new value.

Combine Diverse Strengths.
Understand and respect each other, and bring together our strengths. By learning through challenges and turning failures into the foundation for the future, we overcome ambiguity and difficulties together.

Continue to Evolve.
Take pride in our work and keep evolving with flexibility. Refining our technologies and craftsmanship drives change and leads to further advancement.

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

▶ Human Resources Around the Globe that Support the Electronic Materials Business

Copyright © 2026 Ajinomoto Co., Inc. All rights reserved

Eat Well, Live Well.



AJINOMOTO®